

# P25 J3363: Non-uniform Refinement

CryoSPARC Running Version: v4.7.1

## Project Details

|                 |                                                                                                                      |
|-----------------|----------------------------------------------------------------------------------------------------------------------|
|                 |                                                                                                                      |
| DETAIL          | VALUE                                                                                                                |
| Title           | 24sep20b_WT-Control-White_From-January                                                                               |
| Description     | This is a repeat collection of White Control Wildtype CFTR from the original January/February 2024 collection. Using |
| ID              | P25                                                                                                                  |
| Created by user | Andy Paige                                                                                                           |
|                 |                                                                                                                      |

## Job Details

|                 |                                                           |  |
|-----------------|-----------------------------------------------------------|--|
|                 |                                                           |  |
| DETAIL          | VALUE                                                     |  |
| Title           | New Job J3363                                             |  |
| Description     | Enter a description.                                      |  |
| ID              | J3363                                                     |  |
| Job type        | Non-uniform Refinement                                    |  |
| Status          | completed                                                 |  |
| Created by user | Minou Emmad                                               |  |
| Created         | Fri Jun 05 2026 17:41:59 GMT-0400 (Eastern Daylight Time) |  |
| Queued          | Fri Jun 05 2026 17:42:44 GMT-0400 (Eastern Daylight Time) |  |
| Launched        | Fri Jun 05 2026 17:57:57 GMT-0400 (Eastern Daylight Time) |  |
| Started         | Fri Jun 05 2026 17:58:06 GMT-0400 (Eastern Daylight Time) |  |
| Waiting         | -                                                         |  |
| Killed          | -                                                         |  |
| Completed       | Fri Jun 05 2026 18:06:08 GMT-0400 (Eastern Daylight Time) |  |
| Failed          | -                                                         |  |
| Last accessed   | Wed Jun 10 2026 15:52:04 GMT-0400 (Eastern Daylight Time) |  |
| Size            | 2.69 GB                                                   |  |
|                 |                                                           |  |

## Inputs

### particles

J3357.imported\_particles

|                                 |  |
|---------------------------------|--|
|                                 |  |
| BLOB                            |  |
| J3357.imported_particles.blob.F |  |
|                                 |  |
| CTF                             |  |
| J3357.imported_particles.ctf.F  |  |
|                                 |  |

|              |  |
|--------------|--|
|              |  |
| ALIGNMENTS3D |  |

```
+-----+
| J3357.imported_particles.alignments3D.F |
+-----+
```

**volume**

```
J3360.volume_class_0
+-----+
| MAP |
+-----+
| J3360.volume_class_0.map.F |
+-----+
```

**mask**

No group connections

**Parameters**

**Particle preprocessing**

| PARAMETER                   | VALUE | DEFAULT | SPEC | ADVANCED |
|-----------------------------|-------|---------|------|----------|
| Window dataset (real-space) | True  | X       |      |          |
| Window inner radius         | 0.85  | X       |      | X        |
| Window outer radius         | 0.99  | X       |      | X        |

**Homogeneous Refinement**

| PARAMETER                                      | VALUE | DEFAULT | SPEC | ADVANCED |
|------------------------------------------------|-------|---------|------|----------|
| Symmetry                                       | C1    | X       |      |          |
| Symmetry relaxation method                     | none  | X       |      | X        |
| Do symmetry alignment                          | True  | X       |      | X        |
| Re-estimate greyscale level of input reference | True  | X       |      | X        |
| Number of extra final passes                   | 0     | X       |      | X        |
| Maximum align resolution (A)                   | None  | X       |      |          |
| Initial lowpass resolution (A)                 | 30    | X       |      | X        |
| GSFSC split resolution (A)                     | 20    | X       |      | X        |
| Force re-do GS split                           | False | X       |      | X        |
| Adaptive Marginalization                       | True  | X       |      | X        |
| Non-uniform refine enable                      | True  | X       |      | X        |
| Non-uniform filter order                       | 8     | X       |      | X        |
| Non-uniform AWF                                | 3     | X       |      | X        |
| Enforce non-negativity                         | False | X       |      | X        |
| Window structure in real space                 | True  | X       |      | X        |
| Skip interpolant premult                       | True  | X       |      | X        |
| Ignore DC component                            | True  | X       |      | X        |
| Ignore tilt                                    | False | X       |      | X        |
| Ignore trefoil                                 | False | X       |      | X        |
| Ignore tetra                                   | False | X       |      | X        |
| Use random ordering of particles               | True  | X       |      | X        |

|                                    |       |   |  |   |  |
|------------------------------------|-------|---|--|---|--|
|                                    |       |   |  |   |  |
| Disable auto batchsize             | False | X |  | X |  |
| Batchsize epsilon                  | 0.001 | X |  | X |  |
| Batchsize snrfactor                | 50    | X |  | X |  |
| Reset input per-particle scale     | True  | X |  | X |  |
| Minimize over per-particle scale   | False | X |  | X |  |
| Scale min/use start iter           | 0     | X |  | X |  |
| Noise priorw                       | 50    | X |  | X |  |
| Noise initw                        | 200   | X |  | X |  |
| Noise initial sigma-scale          | 3     | X |  | X |  |
| Initialize noise model from images | False | X |  | X |  |
| Dynamic mask threshold (0-1)       | 0.2   | X |  | X |  |
| Dynamic mask near (A)              | 6     | X |  | X |  |
| Dynamic mask far (A)               | 14    | X |  | X |  |
| Dynamic mask start resolution (A)  | 12    | X |  |   |  |
| Dynamic mask use absolute value    | False | X |  | X |  |
| Show plots from intermediate steps | True  | X |  |   |  |
| GPU batch size of images           | None  | X |  | X |  |
|                                    |       |   |  |   |  |

Defocus Refinement

|                               |       |         |      |          |  |
|-------------------------------|-------|---------|------|----------|--|
|                               |       |         |      |          |  |
| PARAMETER                     | VALUE | DEFAULT | SPEC | ADVANCED |  |
| Optimize per-particle defocus | False | X       |      | X        |  |
| Num. particles to plot        | 3     | X       |      |          |  |
| Minimum Fit Res (A)           | 20    | X       |      |          |  |
| Defocus Search Range (A +/-)  | 2000  | X       |      |          |  |
| GPU batch size of images      | None  | X       |      | X        |  |
|                               |       |         |      |          |  |

Global CTF Refinement

|                               |       |         |      |          |  |
|-------------------------------|-------|---------|------|----------|--|
|                               |       |         |      |          |  |
| PARAMETER                     | VALUE | DEFAULT | SPEC | ADVANCED |  |
| Optimize per-group CTF params | False | X       |      | X        |  |
| Num. groups to plot           | 3     | X       |      |          |  |
| Binning to apply to plots     | 1     | X       |      | X        |  |
| Minimum Fit Res (A)           | 10    | X       |      |          |  |
| Fit Tilt                      | True  | X       |      |          |  |
| Fit Trefoil                   | True  | X       |      |          |  |
| Fit Spherical Aberration      | False | X       |      |          |  |
| Fit Tetrafoil                 | False | X       |      |          |  |
| Fit Anisotropic Mag.          | False | X       |      |          |  |
|                               |       |         |      |          |  |

|                          |      |   |  |   |  |
|--------------------------|------|---|--|---|--|
|                          |      |   |  |   |  |
| GPU batch size of images | None | X |  | X |  |
|                          |      |   |  |   |  |

Ewald Sphere Correction

|                    |          |         |      |          |  |
|--------------------|----------|---------|------|----------|--|
|                    |          |         |      |          |  |
| PARAMETER          | VALUE    | DEFAULT | SPEC | ADVANCED |  |
|                    |          |         |      |          |  |
| Do EWS correction  | False    | X       |      | X        |  |
|                    |          |         |      |          |  |
| EWS curvature sign | negative | X       |      | X        |  |
|                    |          |         |      |          |  |

Random Seeds

|             |            |         |      |          |  |
|-------------|------------|---------|------|----------|--|
|             |            |         |      |          |  |
| PARAMETER   | VALUE      | DEFAULT | SPEC | ADVANCED |  |
|             |            |         |      |          |  |
| Random seed | 1920297531 | X       |      | X        |  |
|             |            |         |      |          |  |

Compute settings

|                              |       |         |      |          |  |
|------------------------------|-------|---------|------|----------|--|
|                              |       |         |      |          |  |
| PARAMETER                    | VALUE | DEFAULT | SPEC | ADVANCED |  |
|                              |       |         |      |          |  |
| Cache particle images on SSD | True  | X       |      |          |  |
|                              |       |         |      |          |  |
| Low-Memory Mode              | False | X       |      |          |  |
|                              |       |         |      |          |  |

Outputs

particles

|                             |  |
|-----------------------------|--|
|                             |  |
| ALIGNMENTS3D                |  |
|                             |  |
| J3363.particle.alignments3D |  |
|                             |  |

|                    |  |
|--------------------|--|
|                    |  |
| CTF                |  |
|                    |  |
| J3363.particle.ctf |  |
|                    |  |

|                     |  |
|---------------------|--|
|                     |  |
| BLOB (PASSTHROUGH)  |  |
|                     |  |
| J3363.particle.blob |  |
|                     |  |

volume

|                   |  |
|-------------------|--|
|                   |  |
| MAP               |  |
|                   |  |
| J3363.volume.blob |  |
|                   |  |

|                   |  |
|-------------------|--|
|                   |  |
| MAP_SHARP         |  |
|                   |  |
| J3363.volume.blob |  |
|                   |  |

|                   |  |
|-------------------|--|
|                   |  |
| MAP_HALF_A        |  |
|                   |  |
| J3363.volume.blob |  |
|                   |  |

|                   |  |
|-------------------|--|
|                   |  |
| MAP_HALF_B        |  |
|                   |  |
| J3363.volume.blob |  |
|                   |  |

|             |  |
|-------------|--|
|             |  |
| MASK_REFINE |  |
|             |  |

|                   |  |
|-------------------|--|
|                   |  |
| J3363.volume.blob |  |
|                   |  |
|                   |  |
| MASK_FSC          |  |
|                   |  |
| J3363.volume.blob |  |
|                   |  |
|                   |  |
| MASK_FSC_AUTO     |  |
|                   |  |
| J3363.volume.blob |  |
|                   |  |
|                   |  |
| PRECISION         |  |
|                   |  |
| J3363.volume.blob |  |
|                   |  |

mask

|                   |  |
|-------------------|--|
|                   |  |
| MASK_REFINE       |  |
|                   |  |
| J3363.volume.blob |  |
|                   |  |

Target

ganges.biology.columbia.edu (node)

```
> [2026-06-05 17:57:57.87 GMT-4] License is valid.
> [2026-06-05 17:57:57.87 GMT-4] Launching job on lane default target ganges.biology.columbia.edu ...
> [2026-06-05 17:57:57.92 GMT-4] Running job on master node hostname ganges.biology.columbia.edu
> [2026-06-05 17:58:06.62 GMT-4] [CPU: 78.9 MB]      [Avail: 305.50 GB] Job J3363 Started
> [2026-06-05 17:58:06.64 GMT-4] [CPU: 78.9 MB]      [Avail: 305.47 GB] Master running v4.7.1, worker running v4.7.1
> [2026-06-05 17:58:06.66 GMT-4] [CPU: 79.2 MB]      [Avail: 305.46 GB] Working in directory:
                                     /mnt/nfs_HuntLabSynologyDS1621CryoEM/hCFTR/Movies/24sep20b/24
                                     sep20b/rawdata/CS-24sep20b-wt-control-white-from-january/J336
                                     3
> [2026-06-05 17:58:06.66 GMT-4] [CPU: 79.2 MB]      [Avail: 305.46 GB] Running on lane default
> [2026-06-05 17:58:06.67 GMT-4] [CPU: 79.2 MB]      [Avail: 305.46 GB] Resources allocated:
> [2026-06-05 17:58:06.67 GMT-4] [CPU: 79.2 MB]      [Avail: 305.46 GB]   Worker:  ganges.biology.columbia.edu
> [2026-06-05 17:58:06.67 GMT-4] [CPU: 79.2 MB]      [Avail: 305.46 GB]   CPU    :  [8, 13, 14, 15]
> [2026-06-05 17:58:06.68 GMT-4] [CPU: 79.2 MB]      [Avail: 305.46 GB]   GPU    :  [2]
> [2026-06-05 17:58:06.68 GMT-4] [CPU: 79.2 MB]      [Avail: 305.46 GB]   RAM    :  [8, 9, 10]
> [2026-06-05 17:58:06.68 GMT-4] [CPU: 79.2 MB]      [Avail: 305.46 GB]   SSD    :  True
> [2026-06-05 17:58:06.69 GMT-4] [CPU: 79.2 MB]      [Avail: 305.46 GB] -----
                                     -
> [2026-06-05 17:58:06.69 GMT-4] [CPU: 79.2 MB]      [Avail: 305.46 GB] Importing job module for job type nonuniform_refine_new...
> [2026-06-05 17:58:18.40 GMT-4] [CPU: 273.6 MB]      [Avail: 304.42 GB] Job ready to run
> [2026-06-05 17:58:18.41 GMT-4] [CPU: 273.6 MB]      [Avail: 304.42 GB] *****
                                     **
> [2026-06-05 17:58:18.64 GMT-4] [CPU: 286.5 MB]      [Avail: 304.19 GB] Using random seed of 1920297531
> [2026-06-05 17:58:18.64 GMT-4] [CPU: 286.5 MB]      [Avail: 304.18 GB] Loading a ParticleStack with 3264 items...
> [2026-06-05 17:58:18.65 GMT-4] [CPU: 286.5 MB]      [Avail: 304.18 GB]

                                     SSD cache ACTIVE at
                                     /scr/instance_ganges.biology.columbia.edu:39001 (200 GB
                                     reserve) (3 TB quota)

                                     Cache usage                                Amount

                                     Total / Usable                3.58 TiB / 2.73 TiB
                                     Used / Free                  2.73 TiB / 58.20 MiB
                                     Hits / Misses              30.95 GiB / 0.00 B
                                     Acquired / Required         30.95 GiB / 0.00 B

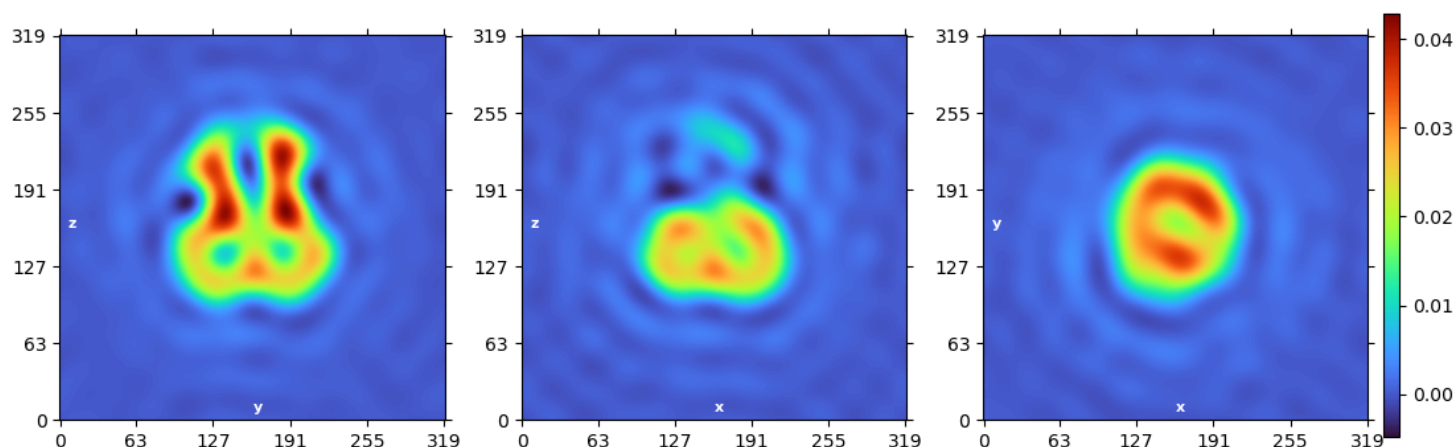
                                     Progress:
                                     [
                                     2657/2657 (100%)
                                     Elapsed: 0h 00m 03s
                                     Active jobs: P25-J3362, P25-J3363, P31-J189
                                     SSD cache complete for 2657 file(s)

> [2026-06-05 17:58:22.08 GMT-4] [CPU: 325.4 MB]      [Avail: 305.46 GB] Done.
> [2026-06-05 17:58:22.08 GMT-4] [CPU: 325.4 MB]      [Avail: 305.46 GB] Windowing particles
> [2026-06-05 17:58:22.09 GMT-4] [CPU: 326.9 MB]      [Avail: 305.45 GB] Done.
> [2026-06-05 17:58:22.10 GMT-4] [CPU: 326.9 MB]      [Avail: 305.46 GB] ===== Gold Standard Split =====
> [2026-06-05 17:58:22.10 GMT-4] [CPU: 326.9 MB]      [Avail: 305.46 GB] Particles have input alignments3D connected, so reusing
                                     pre-existing split
> [2026-06-05 17:58:22.10 GMT-4] [CPU: 326.9 MB]      [Avail: 305.46 GB] Set A is smaller than set B by 8 particles (0.245 percent
                                     difference relative to the total dataset).
> [2026-06-05 17:58:22.12 GMT-4] [CPU: 327.4 MB]      [Avail: 305.45 GB] Split A has 1628 particles
> [2026-06-05 17:58:22.13 GMT-4] [CPU: 327.4 MB]      [Avail: 305.45 GB] Split B has 1636 particles
> [2026-06-05 17:58:22.13 GMT-4] [CPU: 327.4 MB]      [Avail: 305.45 GB] ===== Refinement =====
> [2026-06-05 17:58:22.13 GMT-4] [CPU: 327.4 MB]      [Avail: 305.45 GB] Input particles have box size 320
> [2026-06-05 17:58:22.13 GMT-4] [CPU: 327.4 MB]      [Avail: 305.45 GB] Input particles have pixel size 0.8300
> [2026-06-05 17:58:22.14 GMT-4] [CPU: 327.4 MB]      [Avail: 305.45 GB] Particles will be zeropadded/truncated to size 320 during
                                     alignment
```

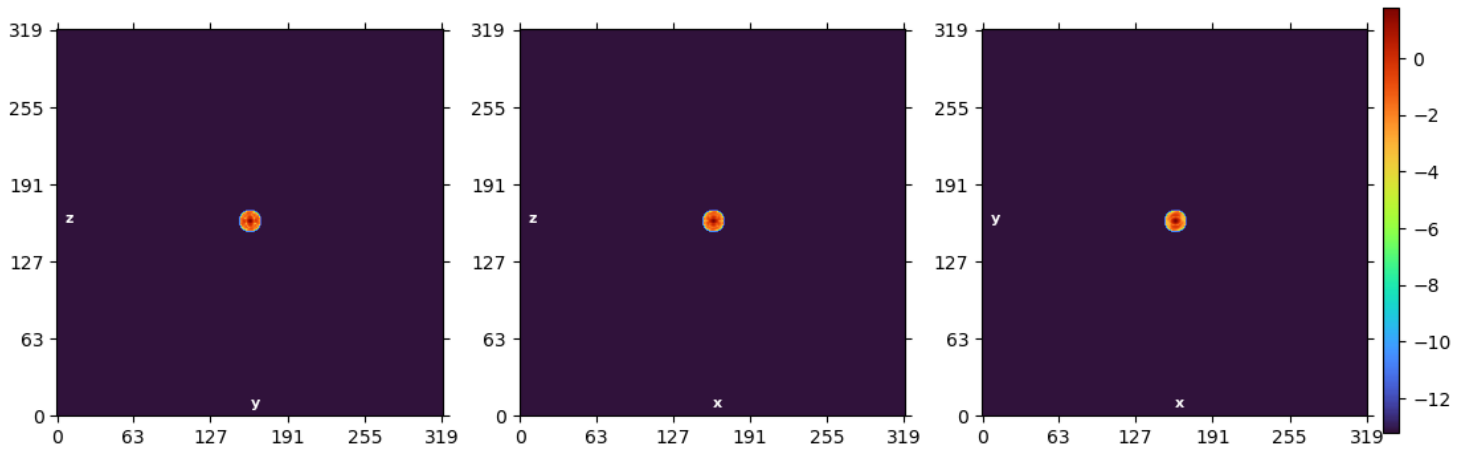
```

> [2026-06-05 17:58:22.14 GMT-4] [CPU: 327.4 MB] [Avail: 305.45 GB] Volume refinement will be done with effective box size 320
> [2026-06-05 17:58:22.14 GMT-4] [CPU: 327.4 MB] [Avail: 305.45 GB] Volume refinement will be done with pixel size 0.8300
> [2026-06-05 17:58:22.15 GMT-4] [CPU: 327.4 MB] [Avail: 305.45 GB] Particles will be zeropadded/truncated to size 320 during
backprojection
> [2026-06-05 17:58:22.15 GMT-4] [CPU: 327.4 MB] [Avail: 305.45 GB] Particles will be backprojected with box size 320
> [2026-06-05 17:58:22.15 GMT-4] [CPU: 327.4 MB] [Avail: 305.45 GB] Volume will be internally cropped and stored with box size
320
> [2026-06-05 17:58:22.16 GMT-4] [CPU: 327.4 MB] [Avail: 305.45 GB] Volume will be interpolated with box size 320 (zeropadding
factor 1.00)
> [2026-06-05 17:58:22.16 GMT-4] [CPU: 327.4 MB] [Avail: 305.45 GB] DC components of images will be ignored and volume will be
floated at each iteration.
> [2026-06-05 17:58:22.16 GMT-4] [CPU: 327.6 MB] [Avail: 305.45 GB] Spherical windowing of maps is enabled
> [2026-06-05 17:58:22.17 GMT-4] [CPU: 327.6 MB] [Avail: 305.45 GB] Refining with C1 symmetry enforced
> [2026-06-05 17:58:22.17 GMT-4] [CPU: 327.6 MB] [Avail: 305.45 GB] Refining with pose and shift marginalization during
backprojection enabled.
> [2026-06-05 17:58:22.17 GMT-4] [CPU: 327.6 MB] [Avail: 305.45 GB] Resetting input per-particle scale factors to 1.0
> [2026-06-05 17:58:22.18 GMT-4] [CPU: 327.6 MB] [Avail: 305.45 GB] Starting at initial resolution 30.000A (radwn 8.853).
> [2026-06-05 17:58:22.18 GMT-4] [CPU: 327.6 MB] [Avail: 305.45 GB] ===== Non-Uniform Refinement =====
> [2026-06-05 17:58:22.18 GMT-4] [CPU: 327.6 MB] [Avail: 305.45 GB] Non-Uniform Refinement is enabled.
> [2026-06-05 17:58:22.18 GMT-4] [CPU: 327.6 MB] [Avail: 305.45 GB] Using AWF of 3.00.
> [2026-06-05 17:58:22.19 GMT-4] [CPU: 327.6 MB] [Avail: 305.45 GB] Using butterworth filter with order 8.
> [2026-06-05 17:58:22.19 GMT-4] [CPU: 327.6 MB] [Avail: 305.45 GB] ===== Masking =====
> [2026-06-05 17:58:24.88 GMT-4] [CPU: 860.2 MB] [Avail: 304.86 GB] No mask input was connected, so dynamic masking will be
enabled.
> [2026-06-05 17:58:24.89 GMT-4] [CPU: 860.2 MB] [Avail: 304.85 GB] Dynamic mask threshold: 0.2000
> [2026-06-05 17:58:24.89 GMT-4] [CPU: 860.2 MB] [Avail: 304.85 GB] Dynamic mask near (A): 6.00
> [2026-06-05 17:58:24.89 GMT-4] [CPU: 860.2 MB] [Avail: 304.85 GB] Dynamic mask far (A): 14.00
> [2026-06-05 17:58:24.90 GMT-4] [CPU: 860.2 MB] [Avail: 304.85 GB] ===== Initial Model =====
> [2026-06-05 17:58:24.90 GMT-4] [CPU: 860.2 MB] [Avail: 304.84 GB] Resampling initial model to specified volume representation
size and pixel-size...
> [2026-06-05 17:58:28.28 GMT-4] [CPU: 1.24 GB] [Avail: 303.48 GB] Estimating scale of initial reference.
> [2026-06-05 17:58:32.93 GMT-4] [CPU: 77.4 MB] [Avail: 302.77 GB] WARNING: io_uring support disabled (not supported by kernel),
I/O performance may degrade
> [2026-06-05 17:58:34.48 GMT-4] [CPU: 1.45 GB] [Avail: 302.15 GB] Rescaling initial reference by a factor of 1.414
> [2026-06-05 17:58:36.16 GMT-4] [CPU: 1.45 GB] [Avail: 302.03 GB] Estimating scale of initial reference.
> [2026-06-05 17:58:38.94 GMT-4] [CPU: 1.41 GB] [Avail: 301.75 GB] Rescaling initial reference by a factor of 1.012
> [2026-06-05 17:58:40.77 GMT-4] [CPU: 1.41 GB] [Avail: 301.63 GB] Estimating scale of initial reference.
> [2026-06-05 17:58:43.66 GMT-4] [CPU: 1.41 GB] [Avail: 301.19 GB] Rescaling initial reference by a factor of 0.993
> [2026-06-05 17:58:46.65 GMT-4] Initial Real Space Slices

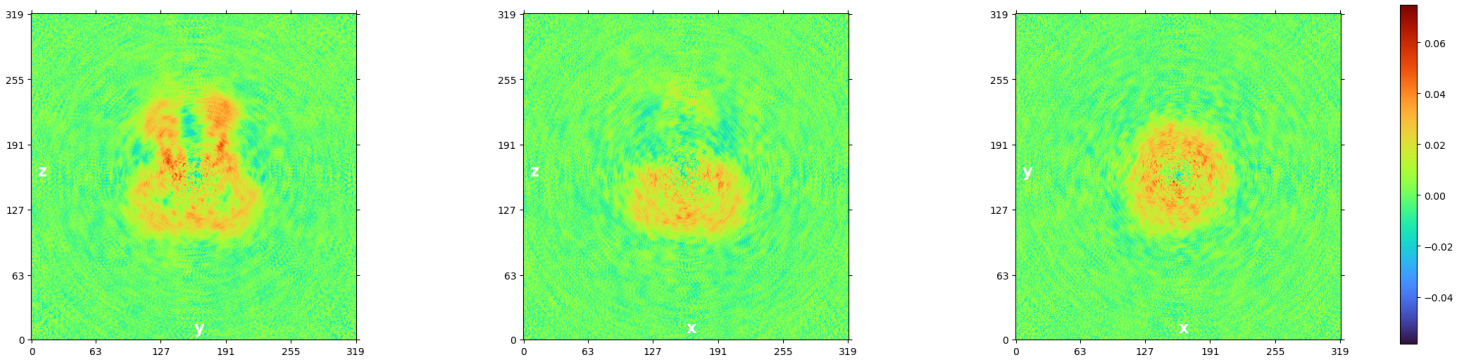
```



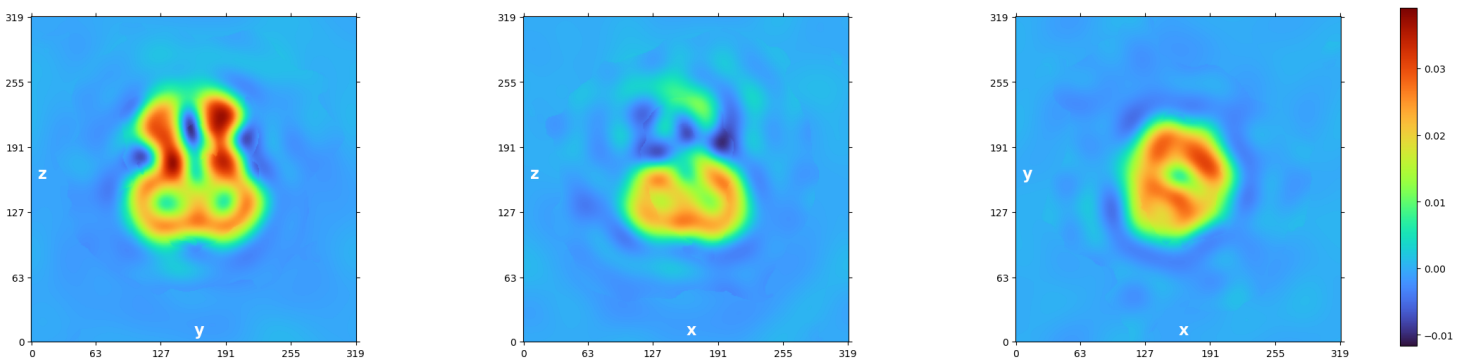
> [2026-06-05 17:58:48.14 GMT-4] Initial Fourier Space Slices



```
> [2026-06-05 17:58:48.14 GMT-4] [CPU: 1.46 GB] [Avail: 301.01 GB] ===== Starting Refinement Iterations =====
> [2026-06-05 17:58:48.15 GMT-4] [CPU: 1.46 GB] [Avail: 301.00 GB] ----- Start Iteration 0
> [2026-06-05 17:58:48.15 GMT-4] [CPU: 1.46 GB] [Avail: 300.99 GB] Using Max Alignment Radius 8.853 (30.000A)
> [2026-06-05 17:58:48.15 GMT-4] [CPU: 1.46 GB] [Avail: 300.98 GB] Auto batchsize: 1636 in each split
> [2026-06-05 17:58:51.92 GMT-4] [CPU: 1.97 GB] [Avail: 300.04 GB] -- THR 1 BATCH 500 NUM 500 TOTAL 1.5136203 ELAPSED 16.833575
--
> [2026-06-05 17:59:11.18 GMT-4] [CPU: 3.19 GB] [Avail: 298.94 GB] Processed 3264.000 images in 19.259s.
> [2026-06-05 17:59:15.05 GMT-4] [CPU: 3.51 GB] [Avail: 296.51 GB] Computing FSCs...
> [2026-06-05 17:59:15.05 GMT-4] [CPU: 3.51 GB] [Avail: 296.50 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 17:59:15.06 GMT-4] [CPU: 3.51 GB] [Avail: 296.49 GB] Computing FFTs on GPU.
> [2026-06-05 17:59:17.55 GMT-4] [CPU: 3.76 GB] [Avail: 295.38 GB] Done in 2.496s
> [2026-06-05 17:59:17.55 GMT-4] [CPU: 3.76 GB] [Avail: 295.38 GB] Computing cFSCs...
> [2026-06-05 17:59:19.42 GMT-4] [CPU: 3.76 GB] [Avail: 295.53 GB] Done in 1.864s
> [2026-06-05 17:59:19.42 GMT-4] [CPU: 3.76 GB] [Avail: 295.53 GB] Using Filter Radius 21.327 (12.454A) | Previous: 8.853
(30.000A)
> [2026-06-05 17:59:25.23 GMT-4] [CPU: 4.83 GB] [Avail: 294.54 GB] Non-uniform regularization with compute option: GPU
> [2026-06-05 17:59:25.23 GMT-4] [CPU: 4.83 GB] [Avail: 294.54 GB] Running local cross validation for A ...
> [2026-06-05 17:59:52.76 GMT-4] [CPU: 7.51 GB] [Avail: 296.11 GB] Local cross validation A done in 27.524s
> [2026-06-05 17:59:54.11 GMT-4] FSC Filtered Side A
```

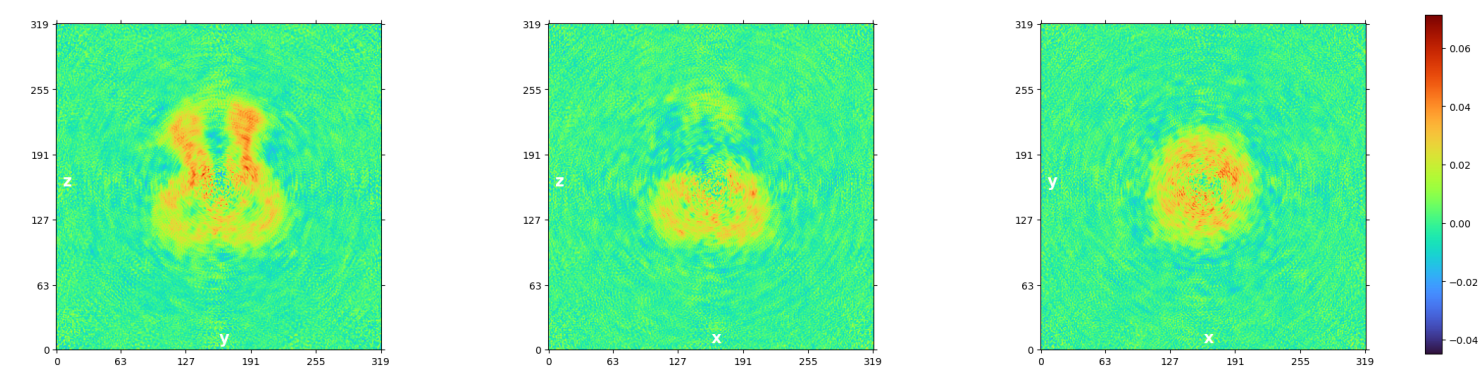


> [2026-06-05 17:59:55.13 GMT-4] CV Filtered Side A

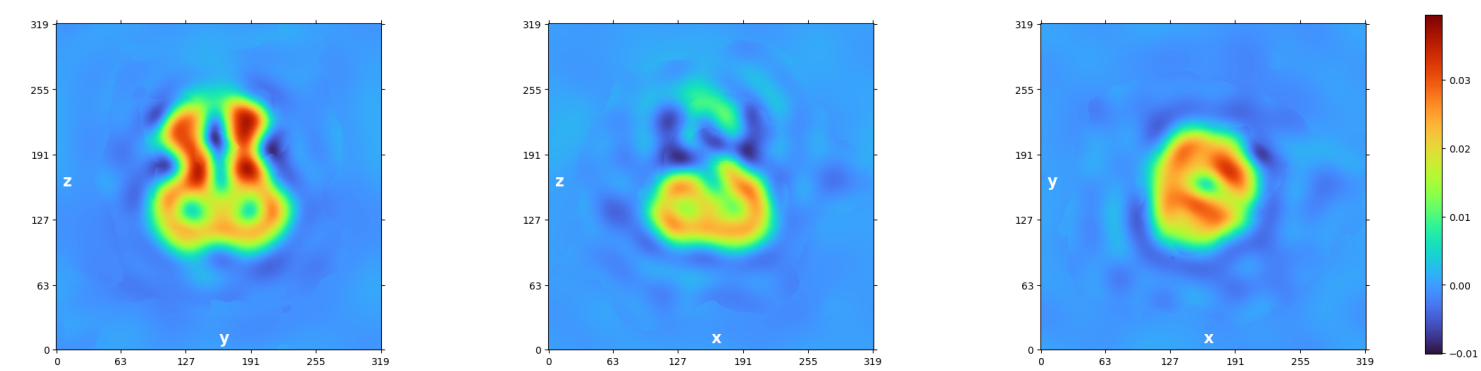


```
> [2026-06-05 17:59:55.14 GMT-4] [CPU: 7.52 GB] [Avail: 296.15 GB] Running local cross validation for B ...
> [2026-06-05 18:00:16.46 GMT-4] [CPU: 7.80 GB] [Avail: 290.94 GB] Local cross validation B done in 21.318s
```

> [2026-06-05 18:00:17.71 GMT-4] FSC Filtered Side B

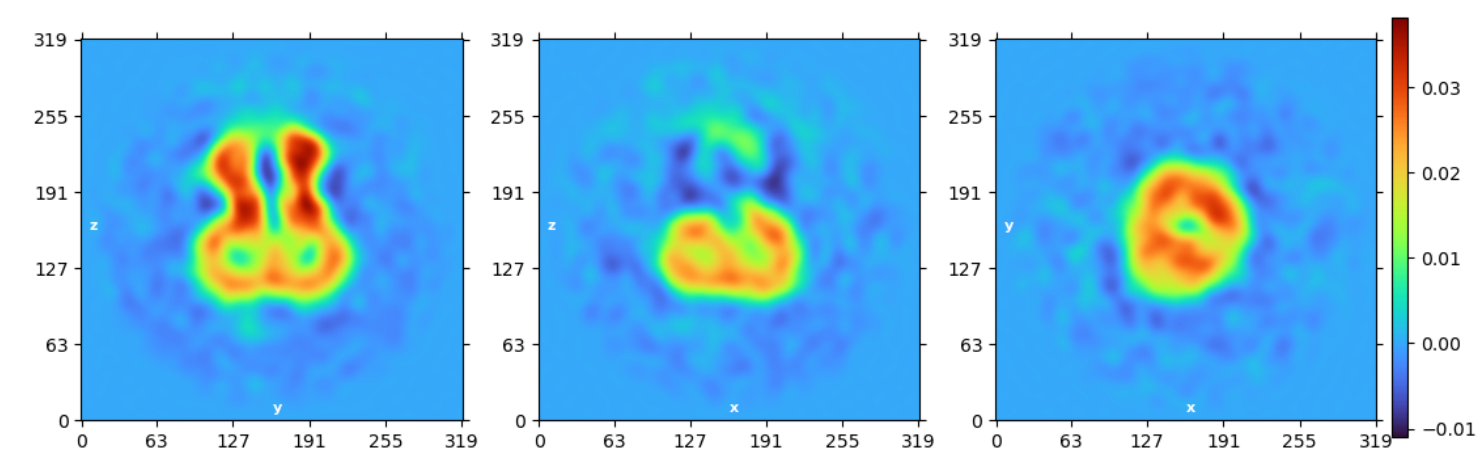


> [2026-06-05 18:00:18.79 GMT-4] CV Filtered Side B

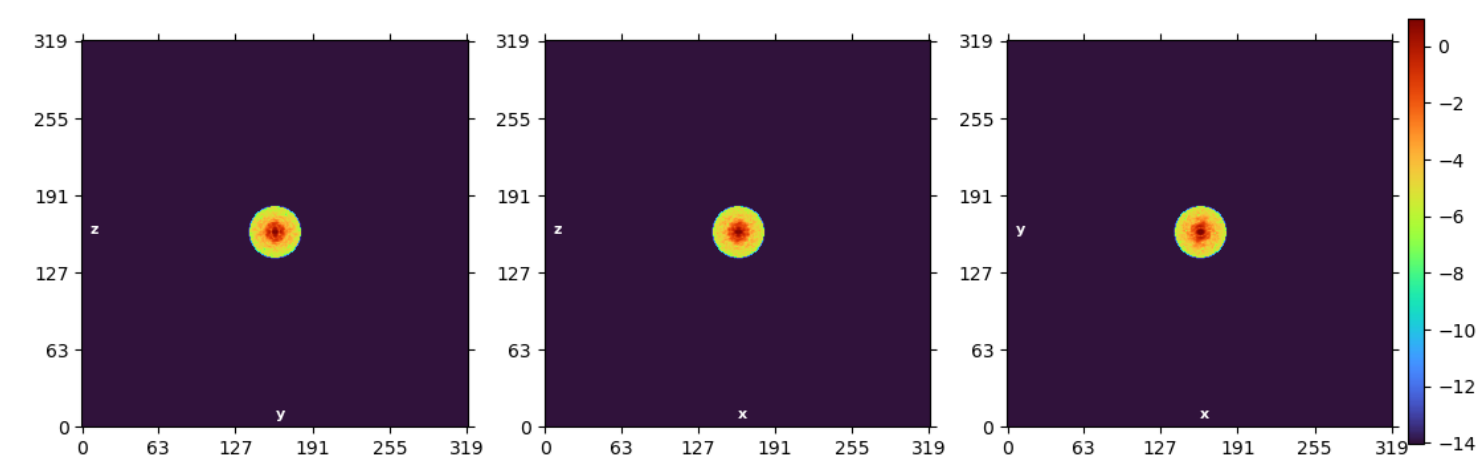


> [2026-06-05 18:00:24.69 GMT-4] [CPU: 8.05 GB] [Avail: 290.26 GB] Plotting..

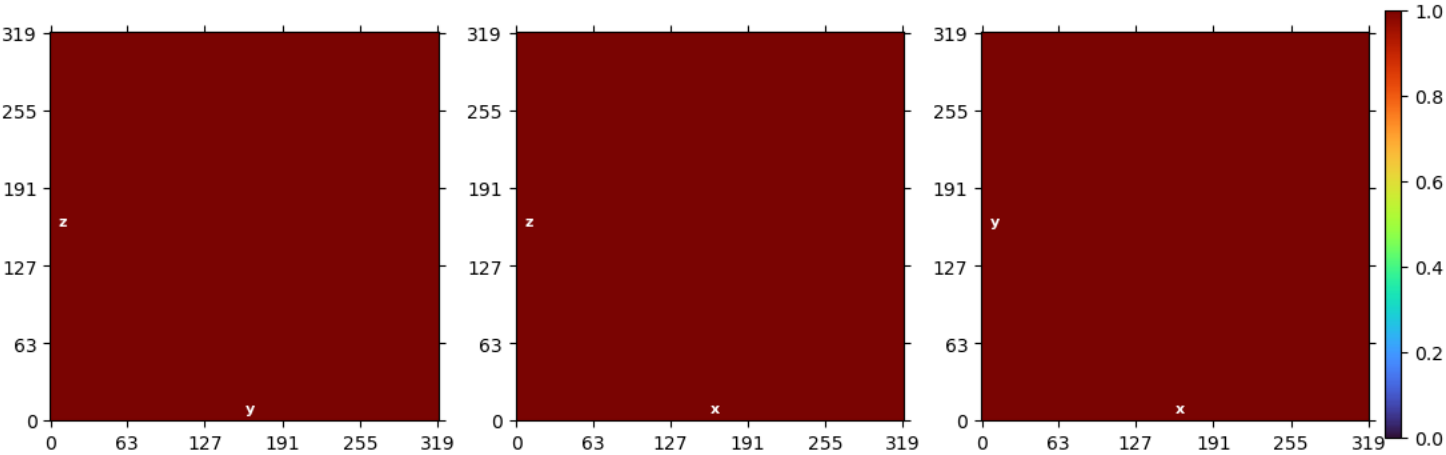
> [2026-06-05 18:00:31.24 GMT-4] Real Space Slices Iteration 000



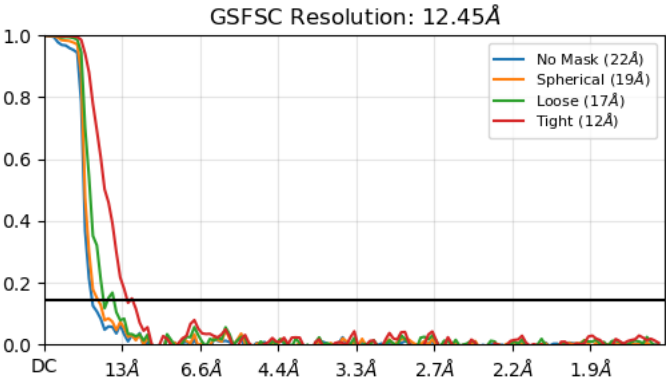
> [2026-06-05 18:00:32.45 GMT-4] Fourier Space Slices Iteration 000



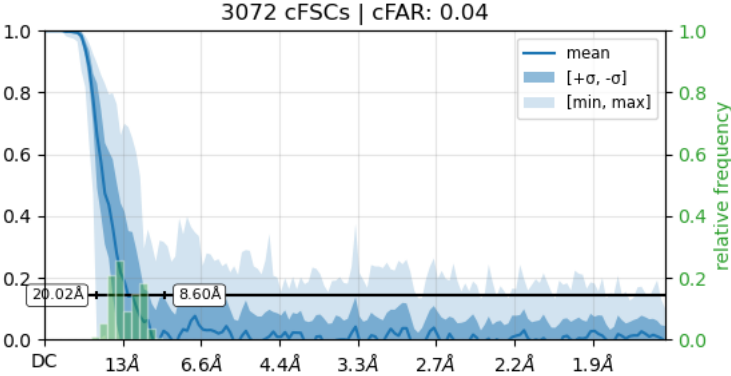
> [2026-06-05 18:00:33.36 GMT-4] Real Space Mask Slices Iteration 000



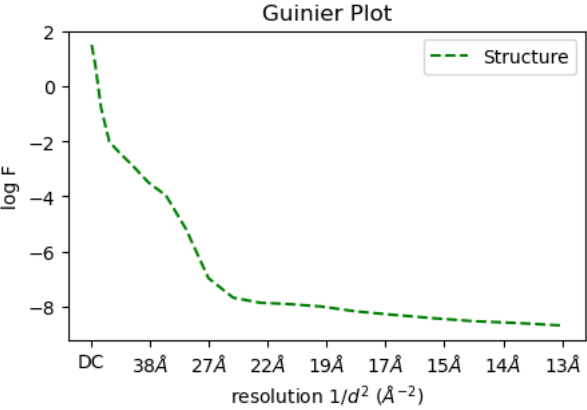
> [2026-06-05 18:00:33.91 GMT-4] FSC Iteration 000



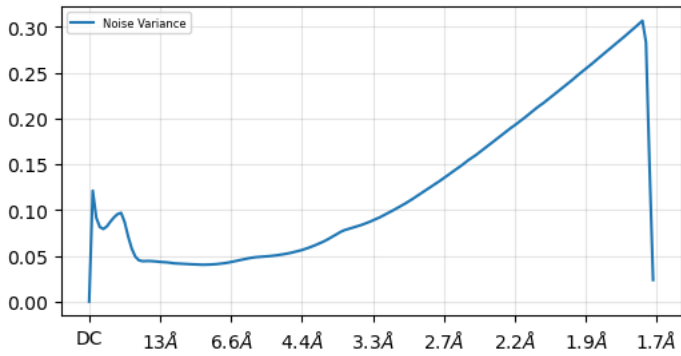
> [2026-06-05 18:00:35.01 GMT-4] cFSCs (Half-angle: 20°) Iteration 000, with tight mask



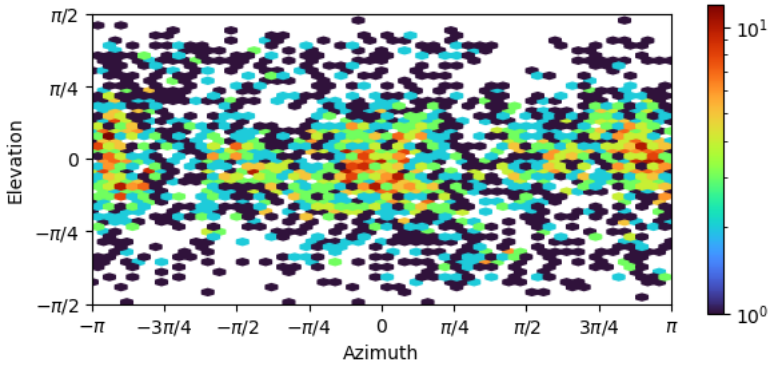
> [2026-06-05 18:00:40.58 GMT-4] Guinier Plot Iteration 000



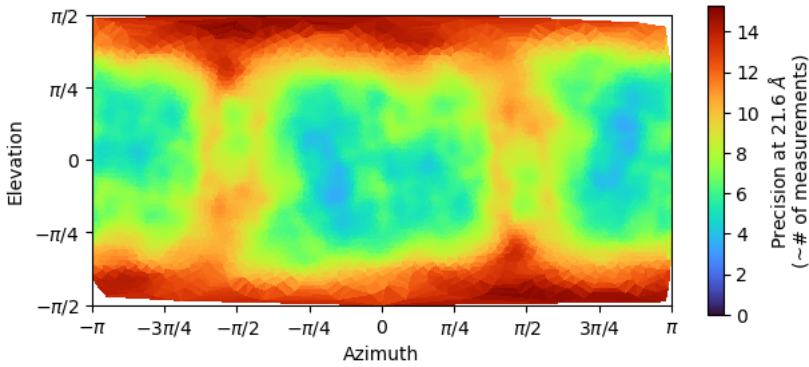
> [2026-06-05 18:00:40.95 GMT-4] Noise Model Iteration 000



> [2026-06-05 18:00:41.67 GMT-4] Viewing Direction Distribution Iteration 000

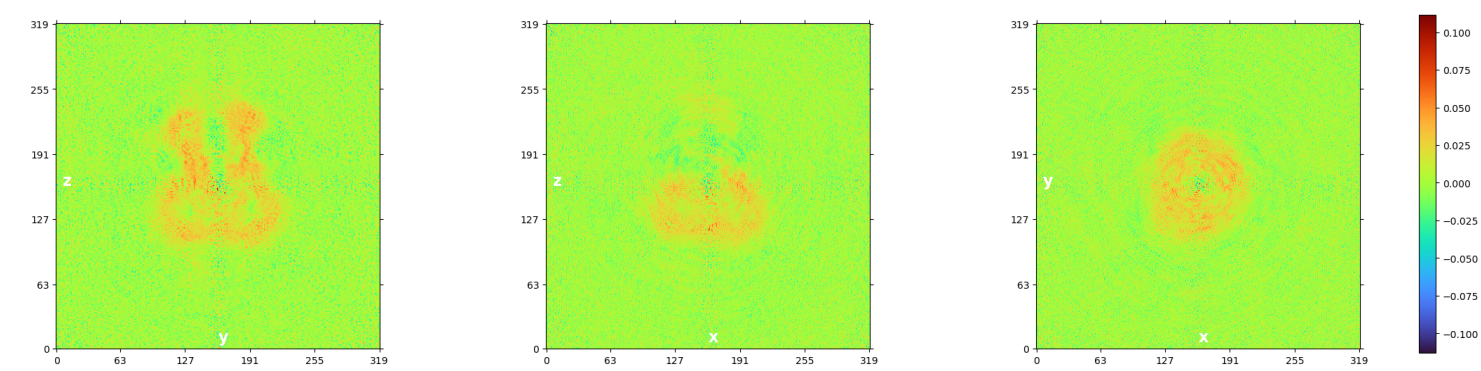


> [2026-06-05 18:00:43.24 GMT-4] Posterior Precision Directional Distribution Iteration 000

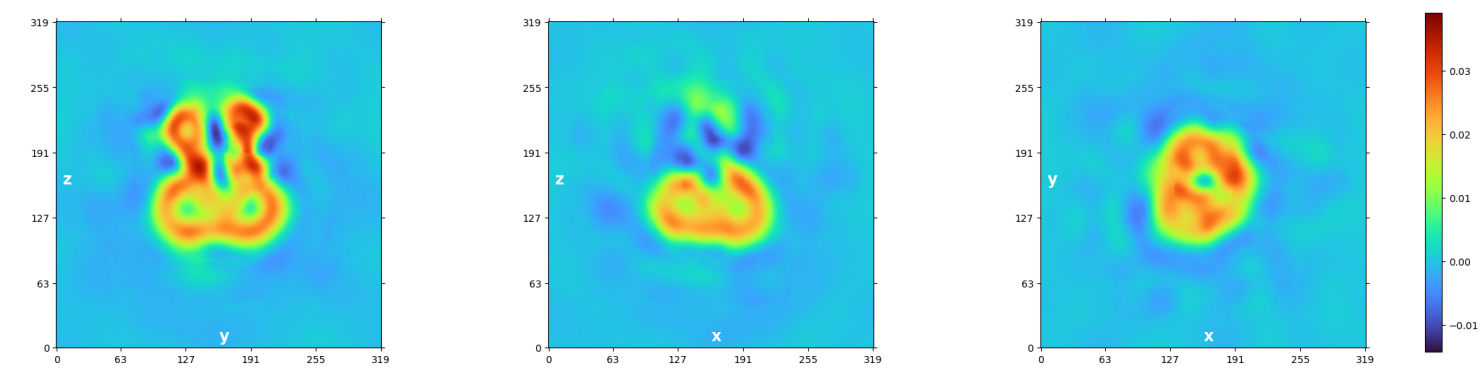


```
> [2026-06-05 18:00:43.24 GMT-4] [CPU: 8.48 GB] [Avail: 291.13 GB] Done in 18.546s.
> [2026-06-05 18:00:43.24 GMT-4] [CPU: 8.48 GB] [Avail: 291.13 GB] Outputting files..
> [2026-06-05 18:00:57.23 GMT-4] [CPU: 8.92 GB] [Avail: 290.01 GB] Done in 13.988s.
> [2026-06-05 18:00:57.24 GMT-4] [CPU: 8.92 GB] [Avail: 290.01 GB] Done iteration 0 in 129.088s. Total time so far 158.801s
> [2026-06-05 18:00:57.80 GMT-4] [CPU: 8.92 GB] [Avail: 290.44 GB] ----- Start Iteration 1
> [2026-06-05 18:00:57.81 GMT-4] [CPU: 8.92 GB] [Avail: 290.44 GB] Using Max Alignment Radius 21.327 (12.454A)
> [2026-06-05 18:00:57.81 GMT-4] [CPU: 8.92 GB] [Avail: 290.44 GB] Auto batchsize: 1636 in each split
> [2026-06-05 18:01:01.52 GMT-4] [CPU: 8.92 GB] [Avail: 292.60 GB] -- THR 1 BATCH 500 NUM 500 TOTAL 1.5925943 ELAPSED 14.609668
--
> [2026-06-05 18:01:18.83 GMT-4] [CPU: 9.77 GB] [Avail: 293.41 GB] Processed 3264.000 images in 17.309s.
> [2026-06-05 18:01:22.53 GMT-4] [CPU: 9.80 GB] [Avail: 293.24 GB] Computing FSCs...
> [2026-06-05 18:01:22.53 GMT-4] [CPU: 9.80 GB] [Avail: 293.24 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:01:22.53 GMT-4] [CPU: 9.80 GB] [Avail: 293.24 GB] Computing FFTs on GPU.
> [2026-06-05 18:01:24.50 GMT-4] [CPU: 9.80 GB] [Avail: 293.21 GB] Done in 1.968s
> [2026-06-05 18:01:24.50 GMT-4] [CPU: 9.80 GB] [Avail: 293.21 GB] Computing cFSCs...
> [2026-06-05 18:01:26.86 GMT-4] [CPU: 9.80 GB] [Avail: 293.13 GB] Done in 2.353s
> [2026-06-05 18:01:26.86 GMT-4] [CPU: 9.80 GB] [Avail: 293.13 GB] Using Filter Radius 25.114 (10.576A) | Previous: 21.327
(12.454A)
> [2026-06-05 18:01:33.38 GMT-4] [CPU: 9.05 GB] [Avail: 296.03 GB] Non-uniform regularization with compute option: GPU
> [2026-06-05 18:01:33.39 GMT-4] [CPU: 9.05 GB] [Avail: 296.03 GB] Running local cross validation for A ...
> [2026-06-05 18:01:57.42 GMT-4] [CPU: 9.27 GB] [Avail: 290.23 GB] Local cross validation A done in 24.026s
```

> [2026-06-05 18:01:58.81 GMT-4] FSC Filtered Side A



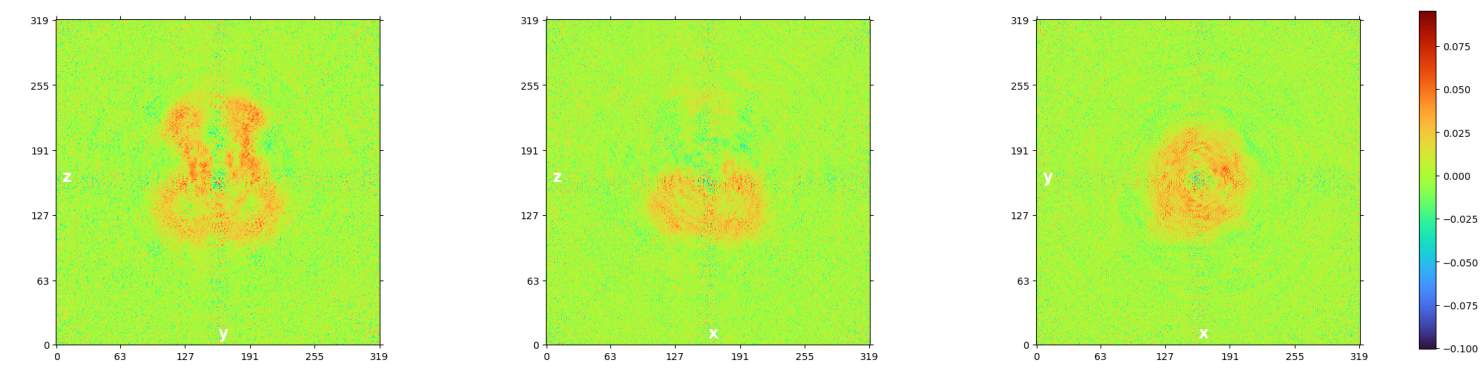
> [2026-06-05 18:01:59.77 GMT-4] CV Filtered Side A



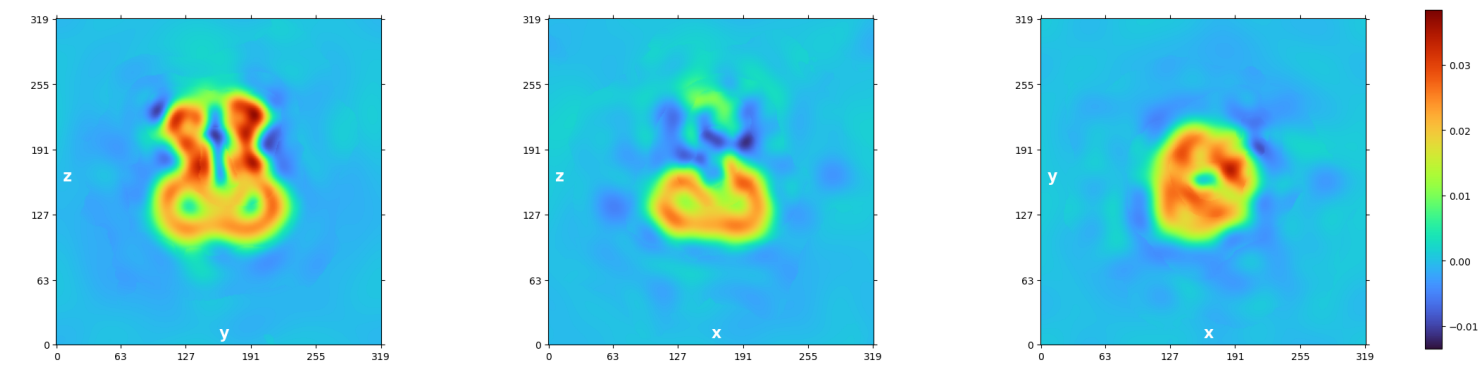
> [2026-06-05 18:01:59.78 GMT-4] [CPU: 9.27 GB] [Avail: 291.71 GB] Running local cross validation for B ...

> [2026-06-05 18:02:22.35 GMT-4] [CPU: 9.52 GB] [Avail: 290.55 GB] Local cross validation B done in 22.567s

> [2026-06-05 18:02:23.67 GMT-4] FSC Filtered Side B

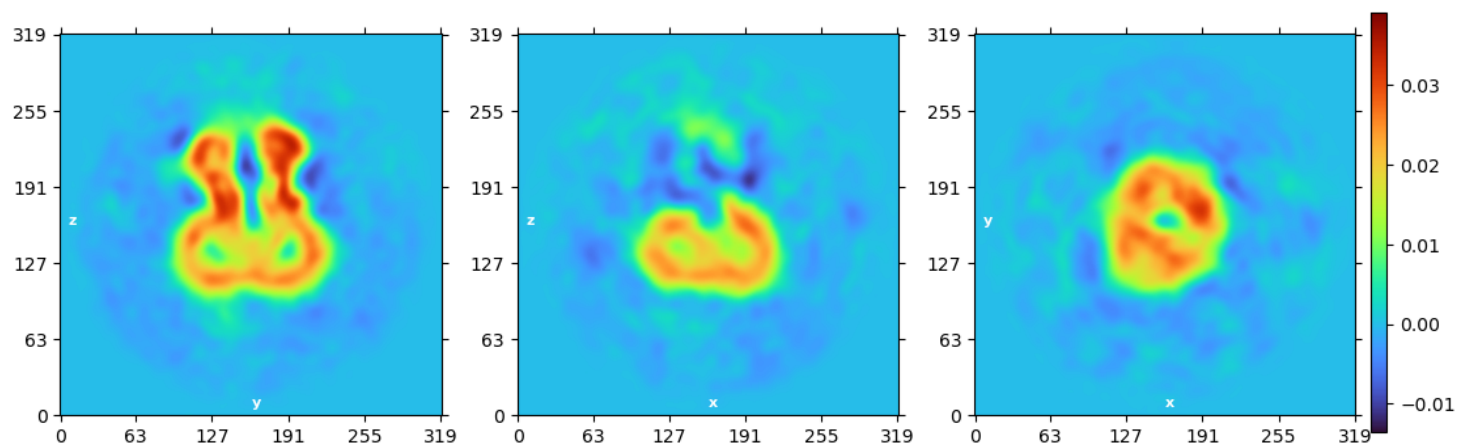


> [2026-06-05 18:02:24.72 GMT-4] CV Filtered Side B

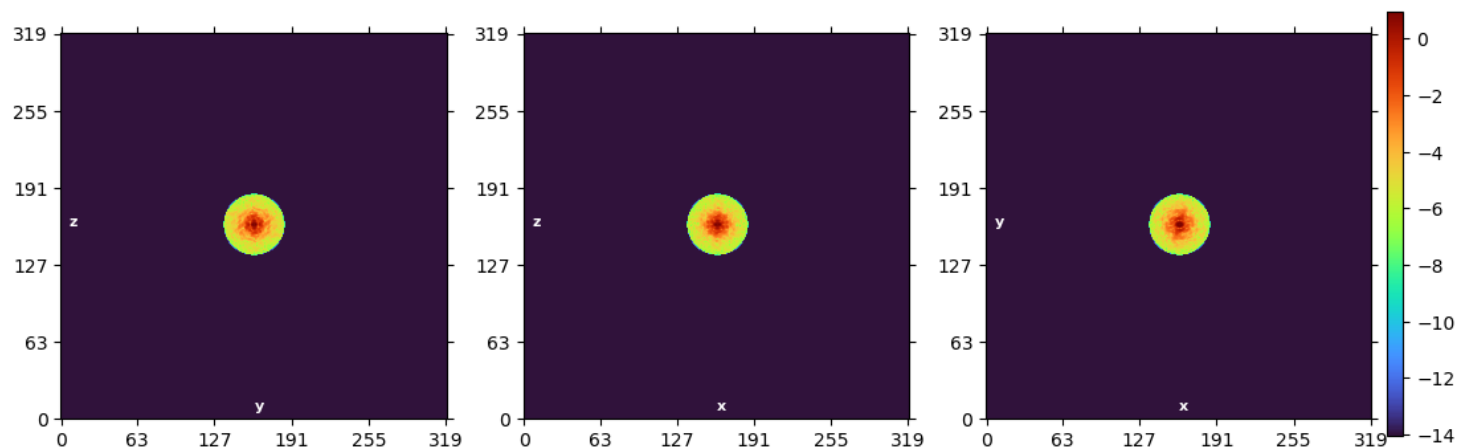


> [2026-06-05 18:02:30.22 GMT-4] [CPU: 9.39 GB] [Avail: 291.29 GB] Plotting..

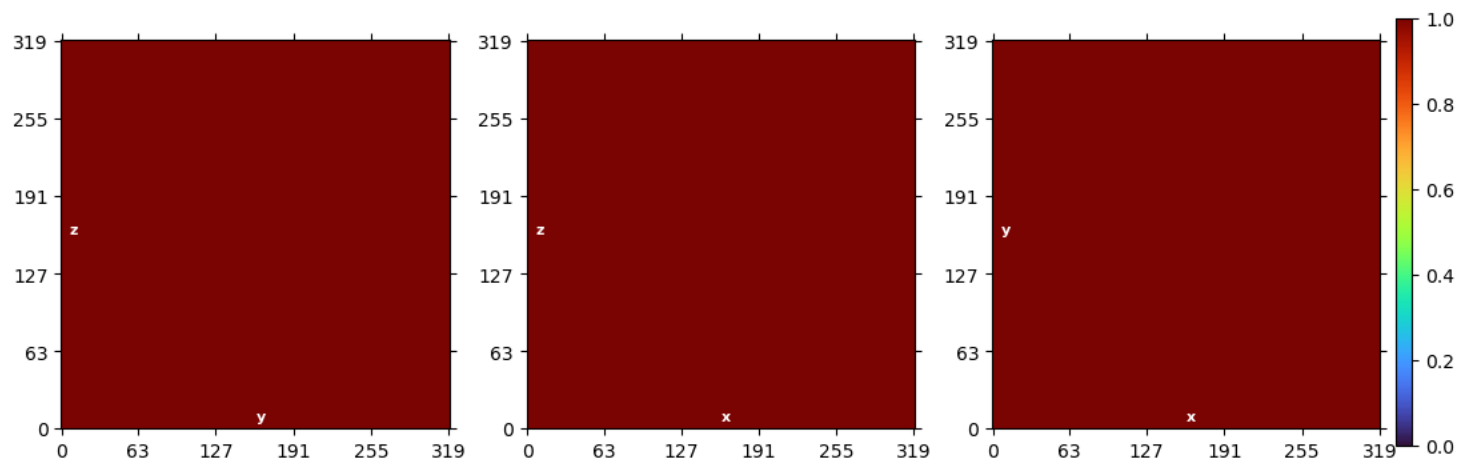
> [2026-06-05 18:02:36.72 GMT-4] Real Space Slices Iteration 001



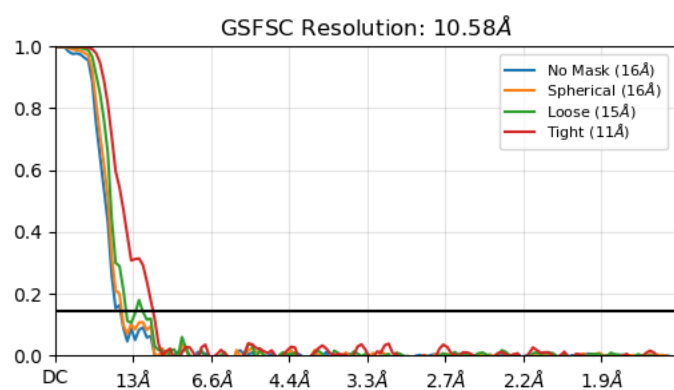
> [2026-06-05 18:02:38.06 GMT-4] Fourier Space Slices Iteration 001



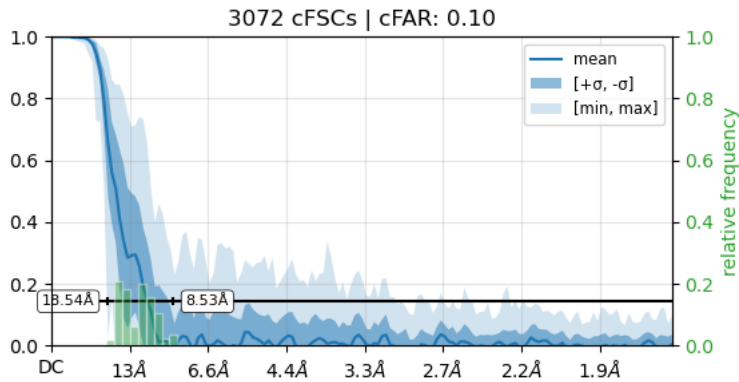
> [2026-06-05 18:02:39.15 GMT-4] Real Space Mask Slices Iteration 001



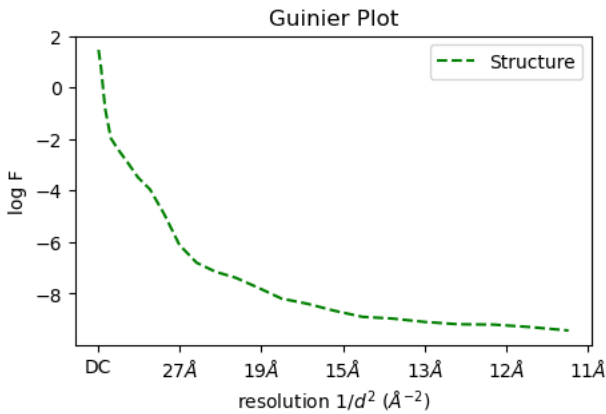
> [2026-06-05 18:02:39.64 GMT-4] FSC Iteration 001



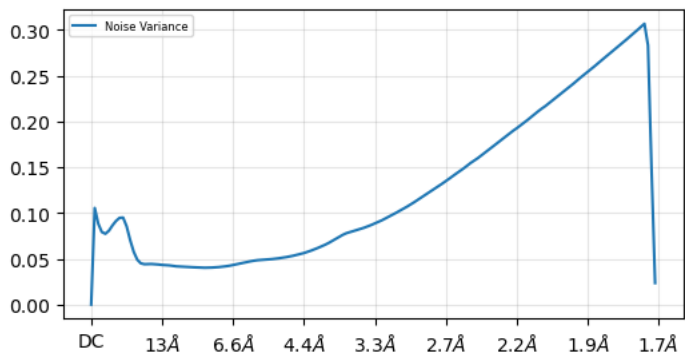
> [2026-06-05 18:02:40.37 GMT-4] cFSCs (Half-angle: 20°) Iteration 001, with tight mask



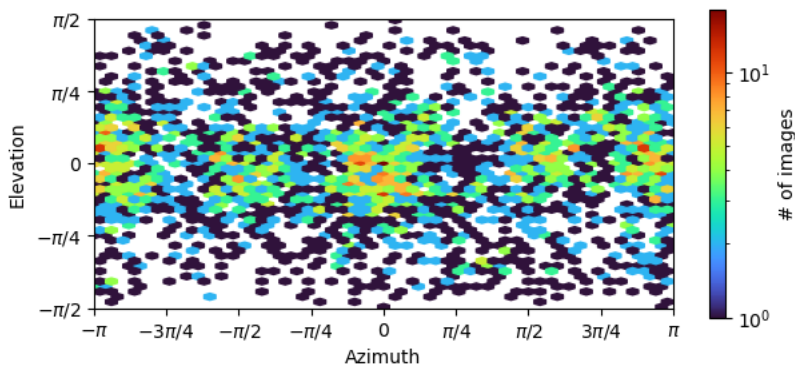
> [2026-06-05 18:02:45.96 GMT-4] Guinier Plot Iteration 001



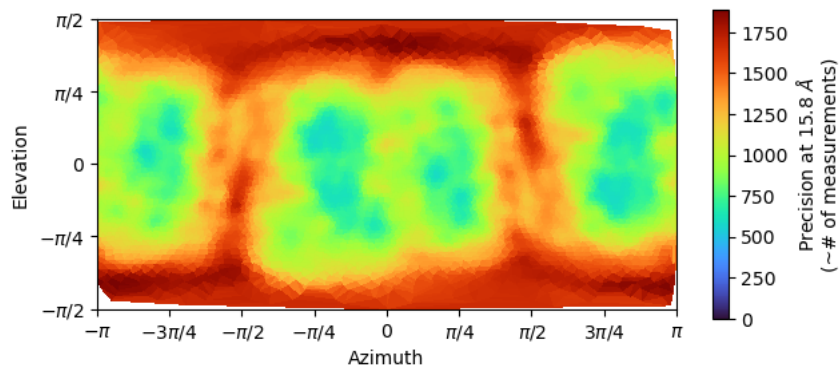
> [2026-06-05 18:02:46.33 GMT-4] Noise Model Iteration 001



> [2026-06-05 18:02:47.08 GMT-4] Viewing Direction Distribution Iteration 001



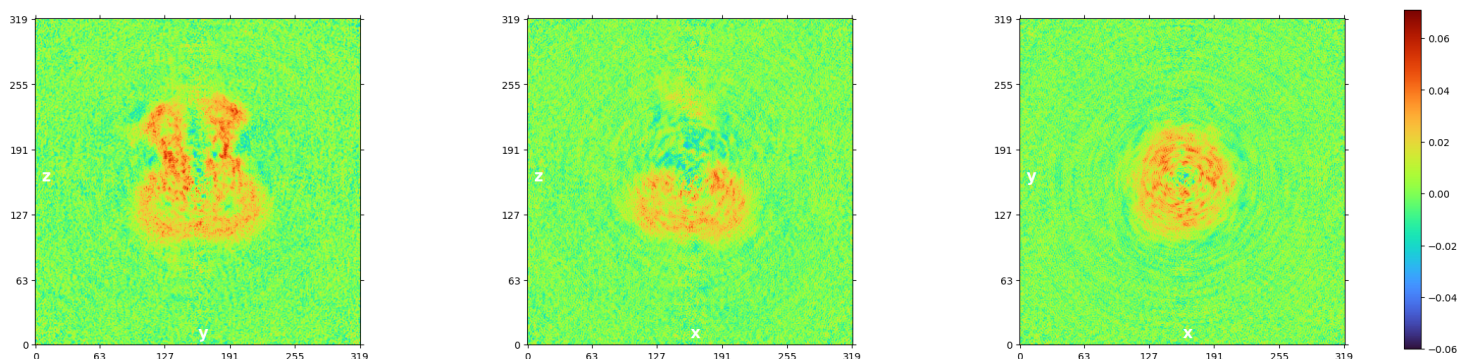
> [2026-06-05 18:02:48.69 GMT-4] Posterior Precision Directional Distribution Iteration 001



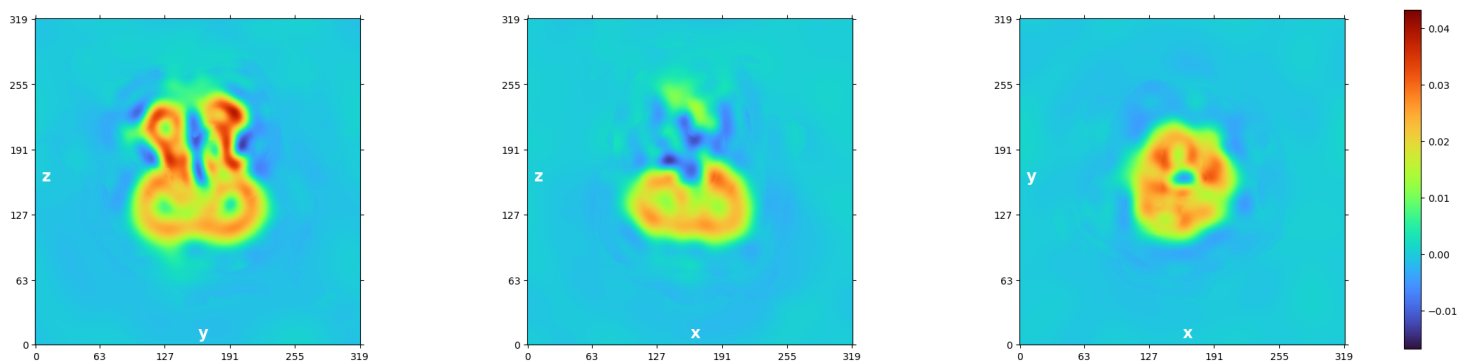
```
> [2026-06-05 18:02:48.69 GMT-4] [CPU: 9.46 GB] [Avail: 296.59 GB] Done in 18.469s.
> [2026-06-05 18:02:48.70 GMT-4] [CPU: 9.46 GB] [Avail: 296.59 GB] Outputting files..
> [2026-06-05 18:03:05.02 GMT-4] [CPU: 9.36 GB] [Avail: 295.26 GB] Done in 16.317s.
> [2026-06-05 18:03:05.02 GMT-4] [CPU: 9.36 GB] [Avail: 295.26 GB] Done iteration 1 in 127.213s. Total time so far 286.582s
> [2026-06-05 18:03:05.74 GMT-4] [CPU: 9.36 GB] [Avail: 295.24 GB] ----- Start Iteration 2
> [2026-06-05 18:03:05.74 GMT-4] [CPU: 9.36 GB] [Avail: 295.24 GB] Using Max Alignment Radius 25.114 (10.576A)
> [2026-06-05 18:03:05.75 GMT-4] [CPU: 9.36 GB] [Avail: 295.24 GB] Auto batchsize: 1636 in each split
> [2026-06-05 18:03:05.75 GMT-4] [CPU: 9.36 GB] [Avail: 295.24 GB] Using dynamic mask.
> [2026-06-05 18:03:33.42 GMT-4] [CPU: 9.86 GB] [Avail: 288.15 GB] -- THR 0 BATCH 500 NUM 500 TOTAL 1.3046214 ELAPSED 14.388509
--
> [2026-06-05 18:03:50.15 GMT-4] [CPU: 10.72 GB] [Avail: 293.07 GB] Processed 3264.000 images in 16.728s.
> [2026-06-05 18:03:53.39 GMT-4] [CPU: 10.75 GB] [Avail: 293.36 GB] Computing FSCs...
> [2026-06-05 18:03:53.40 GMT-4] [CPU: 10.75 GB] [Avail: 293.36 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:03:53.40 GMT-4] [CPU: 10.75 GB] [Avail: 293.36 GB] Computing FFTs on GPU.
> [2026-06-05 18:03:55.41 GMT-4] [CPU: 10.75 GB] [Avail: 294.12 GB] Done in 2.009s
> [2026-06-05 18:03:55.42 GMT-4] [CPU: 10.75 GB] [Avail: 294.19 GB] Optimizing FSC Mask...
> [2026-06-05 18:03:55.44 GMT-4] [CPU: 10.75 GB] [Avail: 294.19 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:03:55.44 GMT-4] [CPU: 10.75 GB] [Avail: 294.19 GB] Computing FFTs on GPU.
> [2026-06-05 18:03:58.83 GMT-4] [CPU: 11.00 GB] [Avail: 293.16 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:03:58.84 GMT-4] [CPU: 11.00 GB] [Avail: 293.16 GB] Computing FFTs on GPU.
> [2026-06-05 18:04:00.93 GMT-4] [CPU: 11.00 GB] [Avail: 291.77 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:04:00.93 GMT-4] [CPU: 11.00 GB] [Avail: 291.76 GB] Computing FFTs on GPU.
> [2026-06-05 18:04:02.91 GMT-4] [CPU: 11.00 GB] [Avail: 292.04 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:04:02.91 GMT-4] [CPU: 11.00 GB] [Avail: 292.04 GB] Computing FFTs on GPU.
> [2026-06-05 18:04:04.88 GMT-4] [CPU: 11.00 GB] [Avail: 292.29 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:04:04.88 GMT-4] [CPU: 11.00 GB] [Avail: 292.29 GB] Computing FFTs on GPU.
> [2026-06-05 18:04:06.79 GMT-4] [CPU: 11.00 GB] [Avail: 292.15 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:04:06.79 GMT-4] [CPU: 11.00 GB] [Avail: 292.15 GB] Computing FFTs on GPU.
> [2026-06-05 18:04:08.74 GMT-4] [CPU: 11.00 GB] [Avail: 291.68 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:04:08.74 GMT-4] [CPU: 11.00 GB] [Avail: 291.72 GB] Computing FFTs on GPU.
> [2026-06-05 18:04:10.70 GMT-4] [CPU: 11.00 GB] [Avail: 291.83 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:04:10.70 GMT-4] [CPU: 11.00 GB] [Avail: 291.83 GB] Computing FFTs on GPU.
> [2026-06-05 18:04:12.64 GMT-4] [CPU: 11.00 GB] [Avail: 292.14 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:04:12.64 GMT-4] [CPU: 11.00 GB] [Avail: 292.13 GB] Computing FFTs on GPU.
> [2026-06-05 18:04:14.69 GMT-4] [CPU: 11.00 GB] [Avail: 296.54 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:04:14.69 GMT-4] [CPU: 11.00 GB] [Avail: 296.54 GB] Computing FFTs on GPU.
> [2026-06-05 18:04:16.71 GMT-4] [CPU: 11.00 GB] [Avail: 297.43 GB] Using full box size 320, downsampled box size 160, with low
```

memory mode disabled.

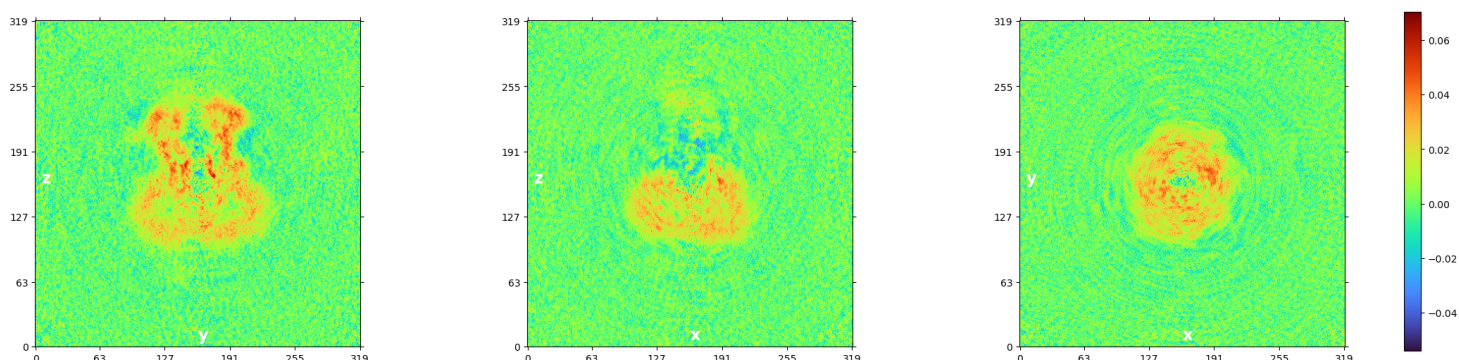
```
> [2026-06-05 18:04:16.72 GMT-4] [CPU: 11.00 GB] [Avail: 297.46 GB] Computing FFTs on GPU.
> [2026-06-05 18:04:18.80 GMT-4] [CPU: 11.00 GB] [Avail: 299.61 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:04:18.80 GMT-4] [CPU: 11.00 GB] [Avail: 299.60 GB] Computing FFTs on GPU.
> [2026-06-05 18:04:20.91 GMT-4] [CPU: 11.00 GB] [Avail: 297.01 GB] Done in 25.482s
> [2026-06-05 18:04:20.91 GMT-4] [CPU: 11.00 GB] [Avail: 297.00 GB] Computing cFSCs...
> [2026-06-05 18:04:22.98 GMT-4] [CPU: 11.01 GB] [Avail: 295.24 GB] Done in 2.060s
> [2026-06-05 18:04:22.98 GMT-4] [CPU: 11.01 GB] [Avail: 295.23 GB] Using Filter Radius 23.591 (11.259A) | Previous: 25.114
(10.576A)
> [2026-06-05 18:04:29.38 GMT-4] [CPU: 10.25 GB] [Avail: 295.00 GB] Non-uniform regularization with compute option: GPU
> [2026-06-05 18:04:29.38 GMT-4] [CPU: 10.25 GB] [Avail: 294.99 GB] Running local cross validation for A ...
> [2026-06-05 18:04:51.68 GMT-4] [CPU: 10.47 GB] [Avail: 303.13 GB] Local cross validation A done in 22.297s
> [2026-06-05 18:04:53.06 GMT-4] FSC Filtered Side A
```



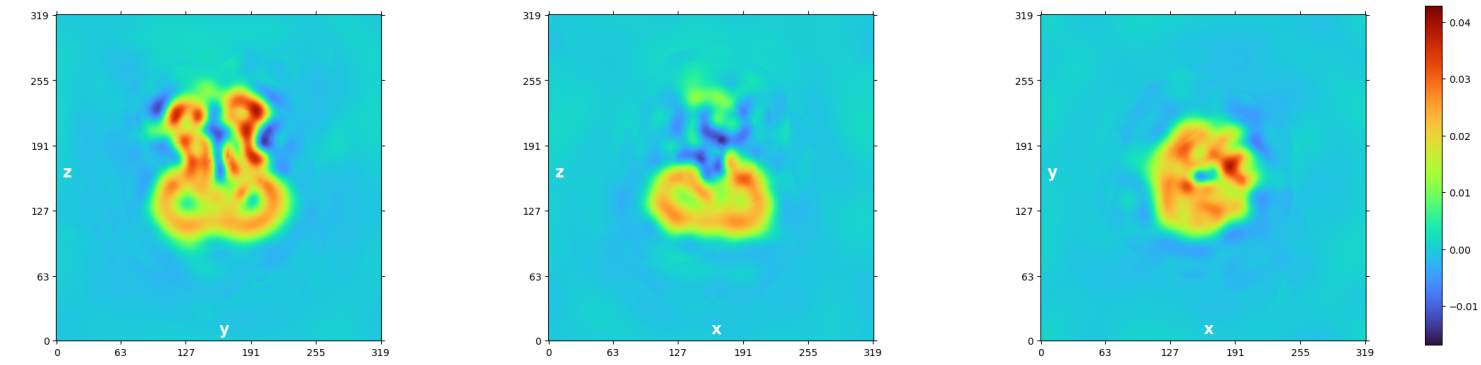
```
> [2026-06-05 18:04:54.10 GMT-4] CV Filtered Side A
```



```
> [2026-06-05 18:04:54.10 GMT-4] [CPU: 10.47 GB] [Avail: 302.62 GB] Running local cross validation for B ...
> [2026-06-05 18:05:16.26 GMT-4] [CPU: 10.72 GB] [Avail: 302.10 GB] Local cross validation B done in 22.157s
> [2026-06-05 18:05:17.62 GMT-4] FSC Filtered Side B
```

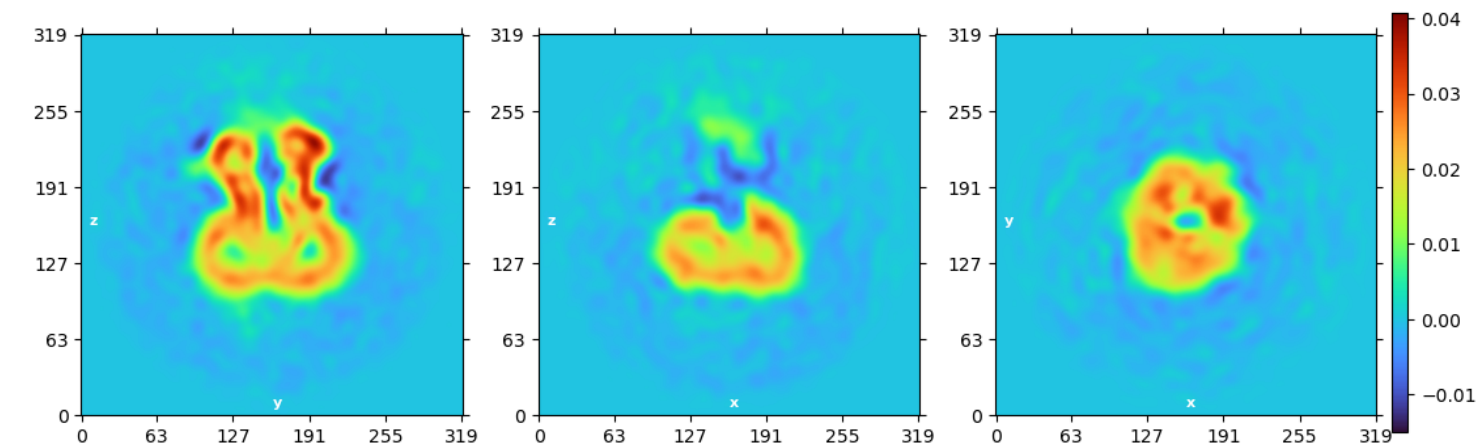


> [2026-06-05 18:05:18.75 GMT-4] CV Filtered Side B

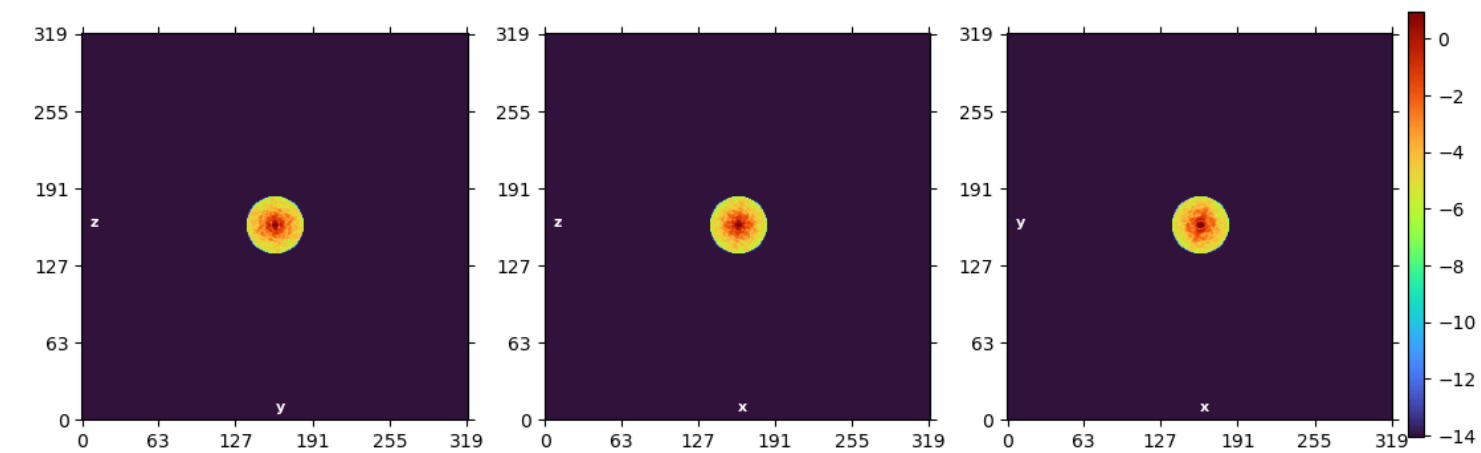


> [2026-06-05 18:05:27.22 GMT-4] [CPU: 10.84 GB] [Avail: 305.11 GB] Plotting..

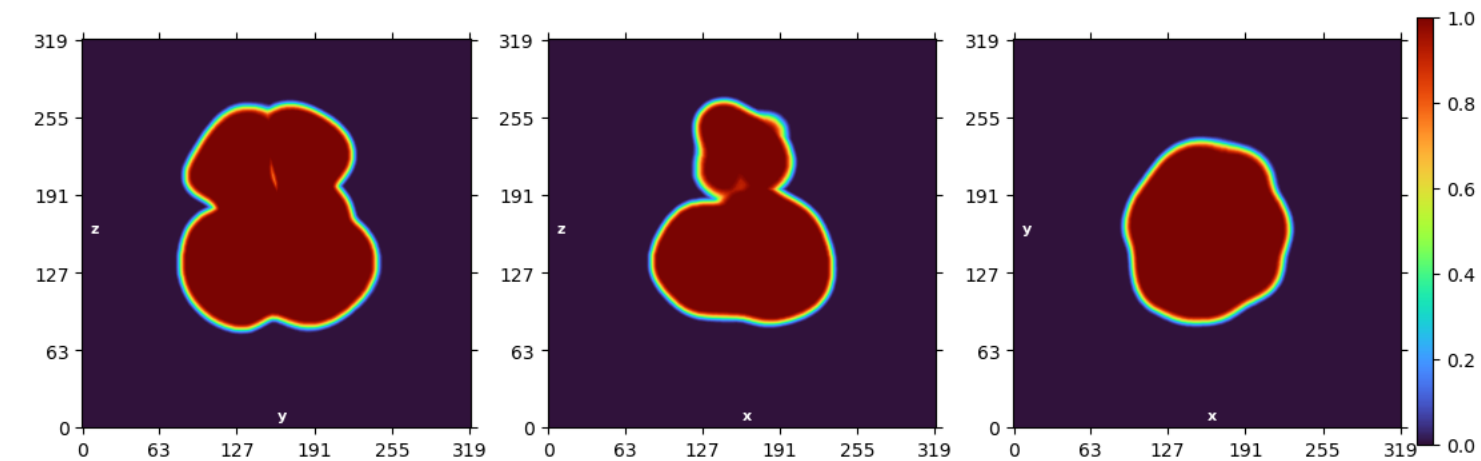
> [2026-06-05 18:05:34.05 GMT-4] Real Space Slices Iteration 002



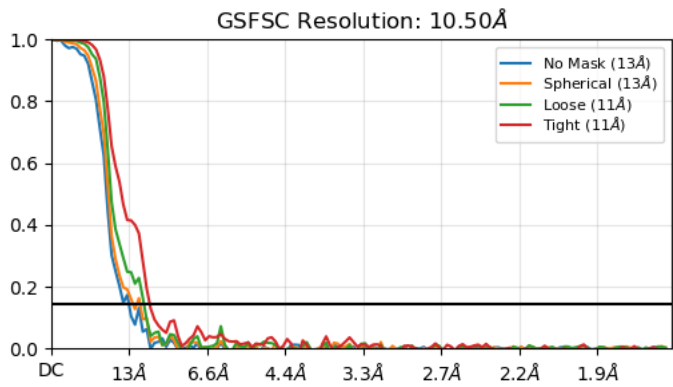
> [2026-06-05 18:05:35.32 GMT-4] Fourier Space Slices Iteration 002



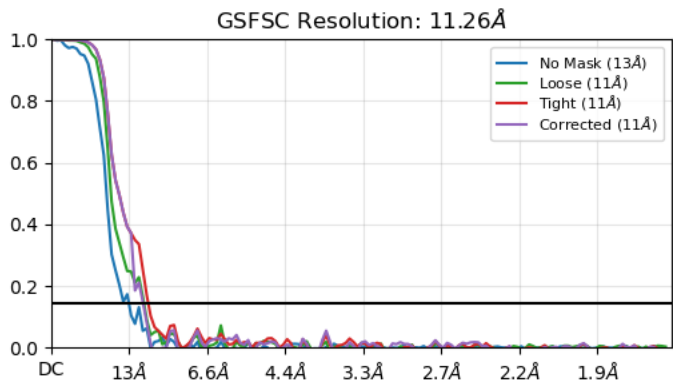
> [2026-06-05 18:05:37.24 GMT-4] Real Space Mask Slices Iteration 002



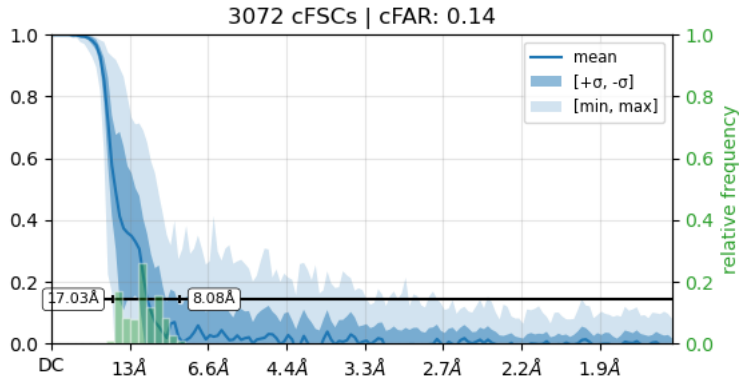
> [2026-06-05 18:05:37.73 GMT-4] FSC Iteration 002



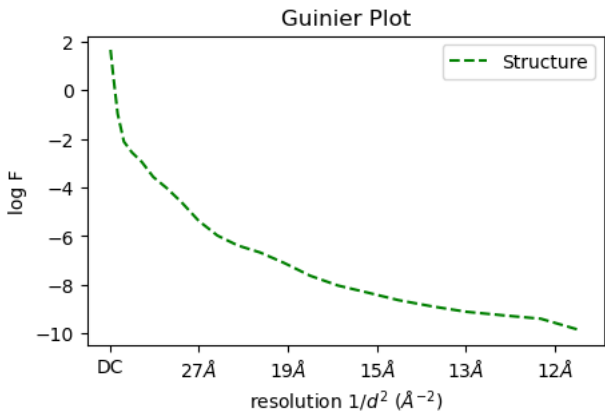
> [2026-06-05 18:05:38.23 GMT-4] FSC Iteration 002, after FSC-mask auto-tightening



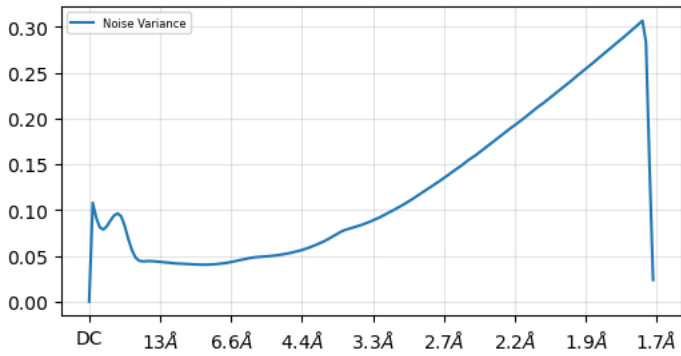
> [2026-06-05 18:05:39.01 GMT-4] cFSCs (Half-angle: 20°) Iteration 002, with autotight mask



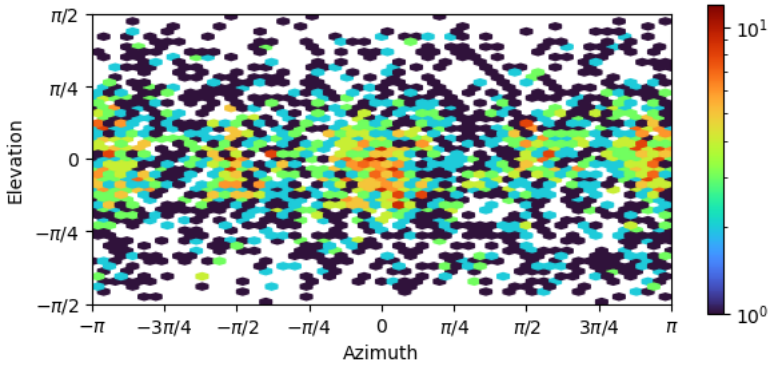
> [2026-06-05 18:05:44.87 GMT-4] Guinier Plot Iteration 002



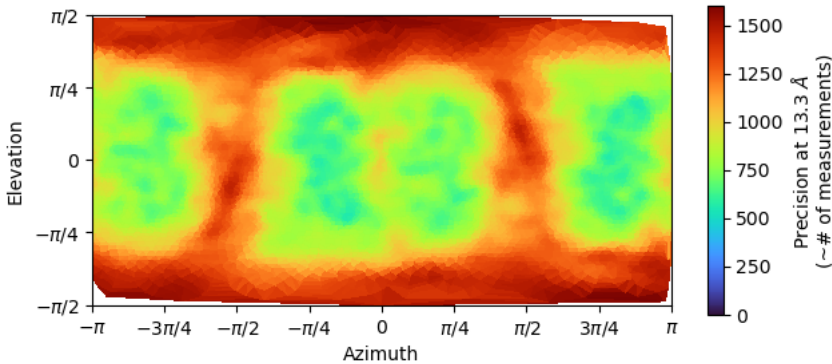
> [2026-06-05 18:05:45.26 GMT-4] Noise Model Iteration 002



> [2026-06-05 18:05:46.09 GMT-4] Viewing Direction Distribution Iteration 002



> [2026-06-05 18:05:48.34 GMT-4] Posterior Precision Directional Distribution Iteration 002



```
> [2026-06-05 18:05:48.34 GMT-4] [CPU: 10.91 GB] [Avail: 302.67 GB] Done in 21.115s.
> [2026-06-05 18:05:48.35 GMT-4] [CPU: 10.91 GB] [Avail: 302.66 GB] Outputting files..
> [2026-06-05 18:06:05.79 GMT-4] [CPU: 10.81 GB] [Avail: 305.04 GB] Done in 17.440s.
> [2026-06-05 18:06:05.79 GMT-4] [CPU: 10.81 GB] [Avail: 305.03 GB] Done iteration 2 in 180.047s. Total time so far 467.355s
> [2026-06-05 18:06:06.60 GMT-4] [CPU: 10.81 GB] [Avail: 303.26 GB] ===== Done Refinement =====
> [2026-06-05 18:06:06.60 GMT-4] [CPU: 10.81 GB] [Avail: 303.28 GB] Note that the output structure from refinement has been
                                                                    lowpass filtered to the gold-standard FSC resolution
                                                                    estimated
                                                                    above. However, the structure probably needs sharpening in
                                                                    order to best visualize high resolution features. This can be
                                                                    done with the Sharpening task (as a new experiment).

> [2026-06-05 18:06:06.61 GMT-4] [CPU: 10.81 GB] [Avail: 303.29 GB] Full run took 468.172s
> [2026-06-05 18:06:07.41 GMT-4] [CPU: 4.95 GB] [Avail: 308.00 GB] -----
                                                                    -

> [2026-06-05 18:06:07.42 GMT-4] [CPU: 4.95 GB] [Avail: 307.99 GB] Compiling job outputs...
> [2026-06-05 18:06:07.42 GMT-4] [CPU: 4.95 GB] [Avail: 307.98 GB] Passing through outputs for output group particles from input
                                                                    group particles

> [2026-06-05 18:06:07.45 GMT-4] [CPU: 4.95 GB] [Avail: 307.95 GB] This job outputted results ['alignments3D', 'ctf']
> [2026-06-05 18:06:07.45 GMT-4] [CPU: 4.95 GB] [Avail: 307.94 GB] Loaded output dataset with 3264 items
> [2026-06-05 18:06:07.46 GMT-4] [CPU: 4.95 GB] [Avail: 307.94 GB] Passthrough results ['blob']
> [2026-06-05 18:06:07.52 GMT-4] [CPU: 4.96 GB] [Avail: 307.86 GB] Loaded passthrough dataset with 3264 items
> [2026-06-05 18:06:07.55 GMT-4] [CPU: 4.96 GB] [Avail: 307.83 GB] Intersection of output and passthrough has 3264 items
> [2026-06-05 18:06:07.57 GMT-4] [CPU: 4.96 GB] [Avail: 307.80 GB] Output dataset contains: ['blob']
```

```
> [2026-06-05 18:06:07.58 GMT-4] [CPU: 4.96 GB] [Avail: 307.80 GB] Outputting passthrough result blob
> [2026-06-05 18:06:07.59 GMT-4] [CPU: 4.96 GB] [Avail: 307.78 GB] Checking outputs for output group particles
> [2026-06-05 18:06:07.67 GMT-4] [CPU: 4.96 GB] [Avail: 307.60 GB] Updating job size...
> [2026-06-05 18:06:07.77 GMT-4] [CPU: 4.96 GB] [Avail: 307.66 GB] Exporting job and creating csg files...
> [2026-06-05 18:06:08.23 GMT-4] [CPU: 4.96 GB] [Avail: 307.78 GB] *****
                                **
> [2026-06-05 18:06:08.24 GMT-4] [CPU: 4.96 GB] [Avail: 307.79 GB] Job complete. Total time 469.84s
```